BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Note: 1. Select the correct Answer:

Q1. If z1 and z2 are any two complex numbers, then:

(a) |z1 + z2| < |z1| + |z2|
(b) |z1 + z2| ≤ |z1| + |z2|
(c) |z1 − z2| < |z1| − |z2|
(d) |z1 + z2| ≥ |z1| + |z2|

Q2. If $z = 3i - 4$, then $z + \bar{z}$ = _____

(a) 8
(b) -3i
(c) -8
(d) (3i - 8)

Q3. If a > 0 and b < 0, then _____

(a) ab > 0
(b) ab < 0
(c) ab = 0
(d) all of them

Q4. If a matrix A has m row and n column, then order of A is:

(a) m x n
(b) n x m
(c) mn
(d) m^n

Q5. Any matrix of order m x 1 is called:

(a) Row matrix
(b) Column matrix
(c) square matrix
(d) zero matrix

Q6. A square matrix A is skew symmetric if:

(a) A^t = A
(b) A^t = -A
(c) (A^t)^t = A
(d) None

Q7. A square matrix A is Hermitian matrix if:

(a) A^t = A
(b) A^t = -A
(c) (Ā)^t = A
(d) (Ā)^t = -A

Q8. Each diagonal element of main diagonal of a skew Hermitian matrix must be:

(a) 1
(b) 0
(c) Any non-zero number
(d) Any complex number

Q9. If |a b; 0 7| = |2 3; 0 -9| = then _____

(a) a = -3
(b) a = b
(c) a = 1/3
(d) a = -31

Q10. The magnitude of $\vec{a}$ x $\vec{b}$ represents the _____ of a parallelogram with adjacent sides $\vec{a}$ and $\vec{b}$.

(a) Opposite sides
(b) diagonal
(c) area
(d) volume

Q11. The volume of tetrahedron determined by vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ is:

(a) [$\vec{a}$, $\vec{b}$, $\vec{c}$]
(b) 1/3[$\vec{a}$, $\vec{b}$, $\vec{c}$]
(c) 1/6[$\vec{a}$, $\vec{b}$, $\vec{c}$]
(d) None

Q12. $\vec{a}$. ($\vec{b}$, $\vec{c}$) = _____

(a) $\vec{a}$. $\vec{b}$, $\vec{c}$
(b) $\vec{a}$ x b x , $\vec{c}$
(c) $\vec{b}$. ($\vec{a}$ x $\vec{c}$)
(d) ($\vec{a}$ x $\vec{b}$) . $\vec{c}$

Q13. If a_n, a_{n+1} = n + 1 and a_4 = 14 then a_5 =

(a) 3
(b) 5
(c) 14
(d) 20

Q14. A sequence {a_n} in which a_{n+1} - a_n is the same number for all n ∈ N is called:

(a) A.P
(b) G.P
(c) H.P
(d) None of these

Q15. If a_{n-1}, a_n, a_{n+1} are in A.P, then a_n is called:

(a) A.M
(b) G.M
(c) H.M
(d) Mid-point

Q16. tan^-1(-x) = _____

(a) -tan^-1 x
(b) π - tan^-1 x
(c) cot^-1 x
(d) tan^-1 x

Q17. Tan^-1(-1) = _____

(a) π/4 (b) -π/4

(c) π/2 (d) π/2

Q18. The product of two conjugate complex number is:

(a) A real number (b) An imaginary number

(c) Always zero (d) Not defined

Section-B

Q2. Show that z = 1 ± i satisfies the equation z^2 − 2z + 2 = 0

Q3. Evaluate: (i) i^10 + (-i)^12 (ii) (-1)^16 + (-i)^16 (iii) (i)^11 (-i)^14

Q4. Expand by means of the binomial theorem: (i) (2x − 3y)^4 (ii) (3x^2 − 2y^3)^5

Q5. Prove that: (i) 3 + 6 + 9 + ……+ 3n = 3/2 n(n + 1) (ii) 2 + 6 + 18 + ….+ 2.3^{n-1} = 3^n - 1

Q6. Expand by means of the binomial theorem: (i) (2x − 3y)^4 (ii) (3x^2 − 2y^3)^5

Q7. Specify the type of each of the following: (i) [0 i; i 0] (ii) [√3; 4; 7]

Q8. Classify the following: (i) Velocity (ii) Mass (iii) Acceleration

Q9. Simplify the following: (i) (-6, 3) (4, -2) (ii) (8, -4) + (-2, 2) (iii) (2, 3) ÷ (4, 5)

Q10. Find the sum of n terms of series: (i) 3n^2 + n + 1 (ii) n^2 + 4n + 1

Q11. solve the following quadratic equation: (i) z^2 + 4z + 5 = 0 (ii) 34x^2 +- 6z = -1 (iii) z^2 + 64 = 1

Q12. Find the transpose of each of the following matrices: (i) [-4 3 6] (ii) [2i 5i -3i; 0 -6i 2i]

Q13. Let A = [3 -4 1; 4 5 7; -2 -3 8], B = [3 4 -1; 2 -3 5] and C = [-5 5; 6 -3; 2 7] [illegible in source scan]

Q14. Classify the following quantities are scalars or vectors (i) Force (ii) Speed (iii) Velocity (iv) Area (v) Work (vi) Volume (vii) weight (viii) Mass

Section-C

Q15. If A = [-2 1 0; -1 4 3; 0 8 5] then find, A^2 − 5A + 4I.

Q16. If f(x) = x|x|, find the values of: (i) f(2) (ii) f(-10) (c) f(1/2)

Q17. Solve the graph of the following: (i) 3x − y ≤ 5 (ii) x − 3y ≥ 5

Q18. Sketch the graph of the following functions: (i) y = 5x + 2 (ii) y = x^2 - 5 (iii) y = x^5